Math 5601: Introduction to Numerical Analysis
Homework assignment 10

Show all relevant work in detail to justify your conclusions. Partial credit depends upon the work you show.

Problem #1: Consider

$$A = \begin{pmatrix} 5 & -2 \\ -4 & 7 \end{pmatrix}.$$

Find $\|A\|_1$, $\|A\|_\infty$, $\|A\|_2$. Verify that $\rho(A) \leq \|A\|_1$, $\rho(A) \leq \|A\|_\infty$, and $\rho(A) \leq \|A\|_2$.

Problem #2: Let A be the $n \times n$ tridiagonal matrix with

$$a_{ij} = \begin{cases} 4, & if\ i = j, \\ -1, & if\ i = j+1\ or\ i = j-1, \\ 0, & otherwise, \end{cases}$$

Prove that the Jacobi method converges for this matrix by using the following theorem from the lecture slides of Chapter 3.

Theorem: Consider the iterative method

$$\vec{x}^{(k+1)} = B\vec{x}^{(k)} + \vec{c},\ k = 0, 1, 2, \cdots$$

The following are equivalent:

- (a) The iterative method is convergent.
- (b) $\rho(B) < 1$.
- (c) There exists a matrix norm $\|\cdot\|$ such that $\|B\| < 1$.

Problem #3: Consider

$$\begin{aligned} x_1 + x_3 &= 0, \\ -x_1 + x_2 &= 0, \\ x_1 + 2x_2 - 3x_3 &= 0. \end{aligned}$$

Explain the observation for the numerical results of Problem #3 of HW9, based on the the following theorem from the lecture slides of Chapter 3.

Theorem: Consider the iterative method

$$\vec{x}^{(k+1)} = B\vec{x}^{(k)} + \vec{c},\ k = 0, 1, 2, \cdots$$

The following are equivalent:

- (a) The iterative method is convergent.
- (b) $\rho(B) < 1$.
- (c) There exists a matrix norm $\|\cdot\|$ such that $\|B\| < 1$.

Math 5601 - Homework 10 - Tyler Glidden

Problem 1 Consider $A = \begin{pmatrix} 5 & -2 \\ 4 & 7 \end{pmatrix}$

$\|A\|_1 = \max((5+4), |-2|+7) = 9$ //
↳ columns 9 9

$\|A\|_\infty = \max((5+|-2|), (4+7)) = 11$ //
rows

$\|A\|_2 = \sqrt{5^2 + (-2)^2 + 4^2 + 7^2} = \sqrt{25+4+16+49} = \sqrt{94} \approx 9.695$ //

$\rho(A) = \sqrt{\max(|\lambda_i|)}$

Char. $\begin{vmatrix} 5-\lambda & -2 \\ 4 & 7-\lambda \end{vmatrix} = (5-\lambda)(7-\lambda) + 8 = \lambda^2 - 12\lambda + 35 + 8$

$\lambda^2 - 12\lambda + 43$

$\lambda = \frac{12 \pm \sqrt{12^2 - 4 \cdot 1 \cdot 43}}{2} = \frac{6 \pm \sqrt{144-172}}{2} = 6 \pm i\sqrt{7}$

Both +

$\rho(A) = \max(|6 \pm i\sqrt{7}|) = \sqrt{6^2 (\pm\sqrt{7})^2} = \sqrt{36+7} = \sqrt{43} \approx 6.55$

$\rho(A) \leq \|A\|_{1,2,\infty}$

$6.55 \leq 9, \ 11, \ 9.695$ ✓

Problem 2

$$\begin{pmatrix} 4 & -1 & 0 & 0 & 0 & \cdots & 0 \\ -1 & 4 & -1 & & & & \\ 0 & -1 & \ddots & \ddots & & & \\ & 0 & \ddots & \ddots & \ddots & & \\ & & & \ddots & \ddots & & \\ & & & & & & 4 \end{pmatrix}$$

Consider

$X^{k+1} = BX^k + \vec{c}$. For Jacobi Method, $\vec{X}^{k+1} = \underbrace{-D^{-1}(L+U)}_{B}\vec{X}^{(k)} + D^{-1}\vec{b}$

$$B = \begin{bmatrix} D^{-1} \\ \begin{pmatrix} -\frac{1}{4} & & & \\ & -1/4 & & \\ & & \ddots & \\ & & & -1/4 \end{pmatrix} \end{bmatrix} \begin{bmatrix} L+U \\ \begin{pmatrix} 0 & -1 & 0 & & \\ -1 & 0 & -1 & & \\ & -1 & \ddots & \ddots & \\ & & \ddots & \ddots & 0 \\ & & & 0 & -1 \\ & & & -1 & 0 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} 0 & +1/4 & & & \\ +1/4 & 0 & +1/4 & & \\ & +1/4 & 0 & +1/4 & \\ & & +1/4 & 0 & \ddots \\ & & & \ddots & \ddots \end{pmatrix} \\ 0 \end{bmatrix}$$

c) $\exists$ a norm, $\|B\|_\infty = \max_{\text{Rows}}(\Sigma \text{rows}) = |+1/4| + |+1/4| = 1/2 < 1$

$\Rightarrow$ The it. method is convergent

Problem 3

Consider $X_1 + X_3 = 0$

$-X_1 + X_2 = 0$

$X_1 + 2X_2 - 3X_3 = 0$

$D = \begin{bmatrix} 1 & 0 & \\ & 1 & \\ 0 & & -3 \end{bmatrix}$ $L = \begin{bmatrix} 0 & & \\ -1 & 0 & \\ 1 & 2 & 0 \end{bmatrix}$ $U = \begin{bmatrix} 0 & 0 & 1 \\ & 0 & 0 \\ 0 & & 0 \end{bmatrix}$

Jacobi

$B = -D^{-1}(L+U)$

$\Rightarrow B = \begin{bmatrix} -1 & & \\ & -1 & 0 \\ & 0 & \end{bmatrix} \frac{1}{3} \left(\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix} \right)$

$B = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 1/3 & 2/3 & 0 \end{bmatrix}$

$\|B\|_1 = \max(\Sigma \text{columns}) = \underline{1\ 1/3}$ //

$\|B\|_\infty = \max(\Sigma \text{rows}) = \underline{1}$ //

$\|B\|_2 = \sqrt{0^2+0^2+(-1)^2+(1)^2+0^2+0^2+(1/3)^2+(2/3)^2} = \sqrt{2\ 5/9} = \sqrt{\frac{23}{9}} \approx \underline{1.598}$ //

There are many other norms, but this is not promising. No Conclusion

$\rho(B)$

$$\begin{vmatrix} -\lambda & 0 & -1 \\ 1 & -\lambda & 0 \\ 1/3 & 2/3 & -\lambda \end{vmatrix} = -\lambda \begin{vmatrix} -\lambda & 0 \\ 2/3 & -\lambda \end{vmatrix} - 0 \begin{vmatrix} & \\ & \end{vmatrix} - 1 \begin{vmatrix} 1 & -\lambda \\ 1/3 & 2/3 \end{vmatrix} = -\lambda(\lambda^2) - 1(2/3 + 1/3\,\lambda) = -\lambda^3 - 1/3\lambda - 2/3$$

$-\lambda^3 - 1/3\lambda - 2/3 = 0$

$\lambda = -0.7474,\ 0.3737 \pm 0.8674i$

$|\lambda_1| = 0.7474$

$|\lambda_2| = \sqrt{0.3737^2 + (0.8674)^2} = 0.9450$

$|\lambda_3| = |\lambda_2| = \sqrt{0.3737^2 + (-0.8674)^2} = 0.9450$

$\rho(B) = \max(|\lambda|) = 0.9450.$

$\rho(B) < 1$ (Barely) so the method is convergent But slowly

Remark: In HW9, P3, I said Jacobi did not converge. I should try a bigger M (iteration max).

Problem 3 Cont.

Gauss-Siedell

$$B = -(L+D)^{-1}U$$

$$-\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 2 & -3 \end{bmatrix}^{-1}\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2/3 & -1/3 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$\rho(B)$

$$\begin{vmatrix} -\lambda & 0 & 1 \\ 0 & -\lambda & 1 \\ 0 & 0 & -1-\lambda \end{vmatrix} = -\lambda\begin{vmatrix} -\lambda & 1 \\ 0 & -1-\lambda \end{vmatrix} - 0\begin{vmatrix} \quad \end{vmatrix} + 0\begin{vmatrix} \quad \end{vmatrix} = -\lambda(-\lambda(-1-\lambda) - 0) = -\lambda(\lambda+\lambda^2) = -\lambda^2 - \lambda^3$$

$$-\lambda^3 - \lambda^2 \overset{\text{set}}{=} 0 \qquad -\lambda^2(\lambda+1) = 0$$

$$\lambda_{1,2} = 0$$

$$\lambda_3 = -1$$

$$\rho(B) = \max(|\lambda|) = 1$$

The G-S is divergent.